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A PALAEOMAGNETIC STUDY OF THE COAST-PARALLEL JURASSIC DYKE SWARM IN SOUTHERN GREENLAND

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Results are reported from palaeomagnetic samples collected in two traverses across the coast-parallel dyke swarm of southern Greenland. This swarm probably resulted as the consequence of initial rifting between Greenland and Labrador, and a reversal of magnetisation has been found which is correlated on the basis of K–Ar age determinations (~168 m.y.) with the Mateke event of the Middle Jurassic (Bajocian). All of fifteen sites show significant grouping of directions after a.f. cleaning; three have anomalous directions of magnetisation while the remainder (nine normal, three reversed) give a combined mean direction of $D = 336°$, $I = 66°$ ($\alpha_{95} = 4.6°$) with a palaeomagnetic pole at 191°E, 72°N. The dykes exhibit the same corelation between polarity and deuteric oxidation state as that found in Tertiary volcanics. There is a systematic change in magnetisation across the dyke swarm in south Greenland from normal to anomalous to reversed directions; this is interpreted as due to lateral migration of the response to the regional stress field with time. The pole position lies in the vicinity of Jurassic poles from North America after closing the Labrador Sea according to the reconstruction of Bullard, Everett and Smith, but the scatter of these latter poles precludes a confirmation of this reconstruction for Middle Jurassic and earlier times.

1. Introduction

The coast-parallel Jurassic dyke swarm in southern Greenland comprises a limited number of basic dykes dipping steeply seawards and confined to south and southwestern coastal regions (Fig. 1). It follows present coastal trends and is generally related to initial rifting between Greenland and Labrador (e.g., Watt, 1969). These dykes have been sampled for palaeomagnetic studies from two traverses across the swarm in the Tugtutôq and Arsuk regions of southern Greenland.

2. Geologic setting and age

The distribution of sampling sites in the two regions is illustrated in Fig. 1. According to Watt (1969) the dykes are gabbroic varying from slightly quartz normative at the northwesterly part of the swarm where they trend S30°E to slightly olivine and nepheline normative in the southern part of the swarm where they trend nearly east–west. Carbonatite-lamprophyres occur in a restricted area as veins or thin dykes parallel to, and cutting the gabbro dykes; Walton (1966) quotes a K–Ar age of 162 ± 5 m.y. for these rocks. Watt (1969) also reports a K–Ar age by Gale of 138 m.y. and Bridgwater (1970) gives a K–Ar (biotite) age for these lamprophyres of 168 ± 5 m.y. In addition Larsen and Møller (1968) quote a K–Ar (biotite) age of 162 ± 5 m.y. while a camptonite sill predating the swarm at Igdlukasik has been dated at 212 ± 5 m.y. (Bridgwater 1970). The weight of the radiometric evidence thus suggests that the dyke swarm is Middle Jurassic in age, and while the location of the swarm is consistent with it representing initial rifting along the line of the later Labrador Sea, intrusion apparently took place more than 80 m.y. before appreciable crustal separation is recognised in this region (Le Pichon et al., 1971; Vogt and Avery, 1974).

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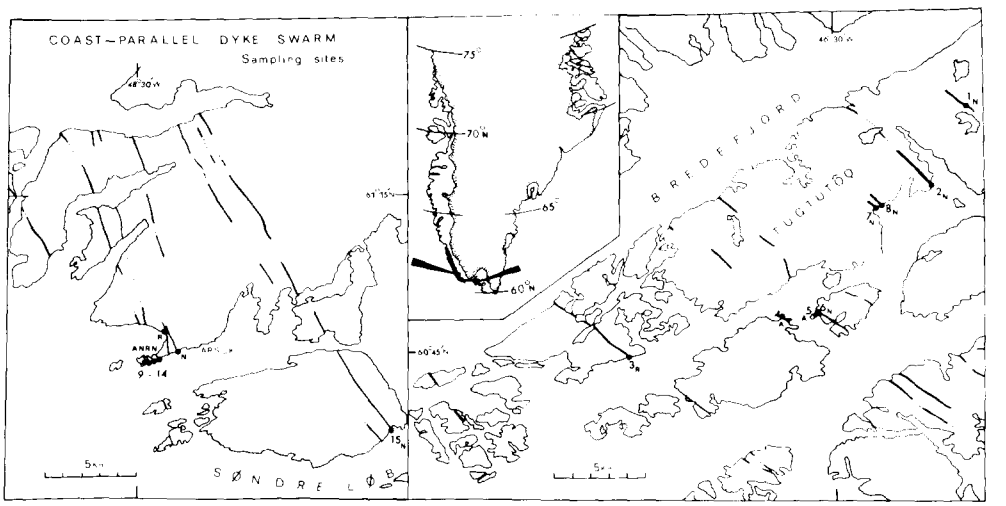


Fig. 1. Distribution and sampling sites of the coast-parallel dykes in southern Greenland. The numbers refer to site numbers of the text and Table I and the small letters at each locality refer to the directions of magnetisation at each site locality as *N* (normal), *R* (reversed), or *A* (anomalous). The inset diagram shows the sampled areas in relation to the outcrop of the coast-parallel dyke swarm in southern Greenland after Watt (1969).

3. Palaeomagnetic results

Samples were collected in the field using a portable motorised drill (Doell and Cox, 1967) and oriented by sun and/or magnetic bearings. One site (*1*) was collected in the form of oriented lumps and cored in the laboratory. Between three and seven separate cores were taken from each dyke and their magnetisations measured in the laboratory using a parastatic magnetometer; samples were treated in demagnetising fields to peak fields between 200 and 700 Oe in steps of 50 or 100 Oe. Behaviour of directions of magnetisation and the decay of magnetic moments with progressive treatment is illustrated in Fig. 2.

On the basis of their magnetic properties samples fall into a poorly stable group (site *4*) showing continuous movement over the whole range of demagnetising fields. A moderately stable group includes samples of sites *5, 11, 12* and *14* with some samples of sites *9* and *10* which show large initial changes in direction followed by little or no change in direction at fields above 100 Oe. All other sites show high stability with little or no movement throughout the range of treatment; a few of these samples do tend to become unstable after treatment in fields higher than 500 Oe. All samples have demagnetisation characteristics typical of titanomagnetite-bearing rocks and have comparable coercivities except for the poorly-stable site *4* which has a large low-coercivity component.

After cleaning all sites yield significantly-grouped directions of magnetisation falling into normal and reversely magnetised groups (Table I). One site (*4*) is remote from either of these groups while site *9* is slightly removed from the remainder of the reversely magnetised group and site *5* is slightly removed from the normally magnetised group; these have been excluded from calculation of the mean direction; sites *4* and *5* have initial moments between 50 and 90% less than the other samples although susceptibilities are comparable (Table II). The twelve remaining sites yield a mean direction of magnetisation $D = 336°$, $I = 66°$ ($k = 90$, $\alpha_{95} = 4.6°$) and a palaeomagnetic pole at 190.9°E, 71.5°N ($A_{95} = 6.8°$).

4. Magnetic properties

High-temperature Curie-point determinations are illustrated in Fig. 2. All thermomagnetic determina-

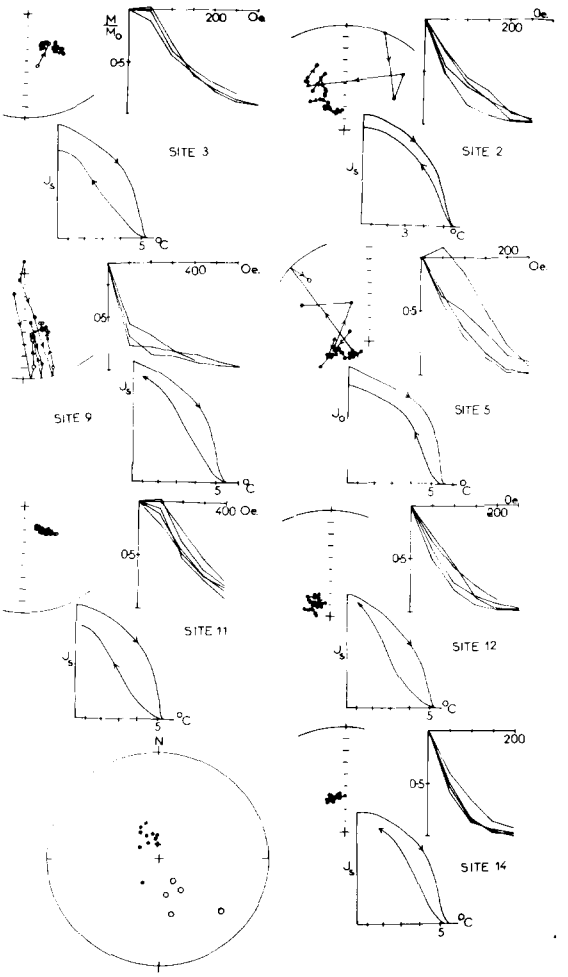


Fig. 2. Behaviour of directions of magnetisation and magnetic moments of typical samples of the dyke swarm in the Tugtutôq and Arsuk regions of southern Greenland. Examples of thermomagnetic (J_S–temperature) curves are also given; the divisions on the abscissa of these graphs represent temperatures in °C times 100. Closed symbols are lower hemisphere (positive inclination) plots and open symbols are upper hemisphere (negative inclination) plots. The bottom left-hand diagram is a plot of the mean directions of all sites from this study; the cross is the direction of the present mean dipole field in this area. M/M_0 is the ratio of the magnetic moment after demagnetisation to the initial moment M_0.

tions show single Curie points between 480 and 570°C typical of partially-oxidised titanomagnetite (Ade-Hall et al., 1965; Larson et al., 1969) with an irreversible change taking place during heatings; in all cases this change produces material with lower saturation magnetisation (J_S). There are several possible causes of this effect (Larson et al., 1969) but since heating took place in air it is inferred to be due to high-temperature oxidation of a metastable phase, probably partial oxidation of titanomagnetite to hematite of lower saturation magnetisation (Nagata, 1961).

Polished-section examination was undertaken on samples of all sites except site *1* which was considerably weathered. Characteristics of the opaque phases are comparable to those reported from other dyke collections (Ade-Hall and Wilson, 1969; Ade-Hall and Lawley, 1970) with low deuteric oxidation states predominating and ubiquitous presence of sulphides (Table II). Following Ade-Hall and Wilson (1969) the deuteric oxidation state was assessed by recording the condition of a number of grains (in this case at least 25) in each section and determining the arithmetic mean. In this collection three oxidation classes were observed ranging from the lowest class (*1*) with no ilmenite exsolution to class *3* with more than 50% of the grain area containing dense ilmenite exsolution. In traverses of each section presence and grain size of separate ilmenite and sulphides were also recorded (Table II).

A correlation observed by other workers (e.g., Wilson and Watkins, 1967; Ade-Hall and Wilson, 1969) between oxidation state and polarity in Tertiary dykes and lavas is also found in this collection: the reversely magnetised dykes all have oxidation states higher than 1.70 while the normally magnetised dykes all have oxidation states below 1.64. The three anomalous sites do not fall in either group but the site with a direction of magnetisation closest to the reversed group (*9*) also has a higher oxidation state (2.50). The reason for this correlation, which is not universal (Ade-Hall and Watkins, 1970), has so far defied satisfactory explanation. Sulphides are only preserved as large grains in samples with low-oxidation states and in general the amount of sulphide decreases with increase in titanomagnetite oxidation state; free ilmenite although rare in this collection is only found in samples with oxidation states higher than 1.64. Maghemite was observed to partially replace grains showing no deuteric oxidation in several samples (Table II) but its presence is not evident in thermomagnetic determinations (Fig. 2).

TABLE I

A.f. cleaned palaeomagnetic results

Site No.	D	I	N	R	k	α_{95}	Virtual geomagnetic pole	
							lat. (°N)	long. (°E)
1	358.8	69.7	3	2.93	28	23.7	82.0	137.3
2	332.0	59.1	7	6.83	35	10.3	62.8	185.9
3	145.5	−57.2	6	5.97	168	5.2	−58.4	12.7
4	131.3	−21.8	4	4.77	13	26.7	−29.0	11.4
5	215.3	65.9	5	4.09	4	41.3	22.8	289.3
6	328.3	74.1	4	3.99	1,240	2.6	74.6	235.6
7	347.2	67.8	6	5.98	250	4.3	77.5	174.5
8	352.2	73.2	5	4.99	391	3.9	85.5	198.5
9	168.3	−40.4	6	5.96	128	5.9	−50.7	329.1
10	334.2	67.7	3	2.99	401	6.1	72.1	196.2
11	148.4	−67.7	6	5.92	68	8.1	−69.6	24.4
12	302.1	71.8	6	5.96	132	5.8	60.8	238.8
13	169.1	−58.1	6	5.96	116	6.2	−66.3	333.5
14	342.6	57.9	6	5.88	42	10.5	64.7	165.1
15	328.2	63.9	6	5.94	84	7.4	65.1	194.5
Mean, excluding sites *4*, *5* and *9* and reversing sites *3*, *11* and *13*:								
			(sites)				(palaeomagnetic pole)	
	336.0	66.2	12	11.88	90	4.6	71.5	190.9
							($d\upsilon$ = 6.2, $d\chi$ = 7.5)	

D and I are declination and inclination of the direction of magnetisation derived from N samples (or sites) with R being the magnitude of the resultant vector and $k = ((N - I)/(N - R))$ the estimate of the precision; α_{95} = circle of 95% confidence of mean direction.

TABLE II

Coast-parallel Jurassic dyke swarm – summary of magnetic and opaque petrologic properties

Site No.	Polarity*	Titanomagnetite oxidation state	Titanomagnetite grain size (μ)	Sulphides grain size (μ) and volume	Ilmenite grain size (volume)	Curie point ±10°C	Mass susceptibility ($\cdot 10^{-3}$)
2	N	1.17	110	375 (16%)	–	530	1.67–1.89
3	R	2.52	16	28 (4%)	23 (10%)	520	1.27–1.39
4	A	1.06	100	24 (10%)	–	480	1.20–2.00
5	A	1.12**	80	18 (14%)	–	560	0.85–1.16
6	N	1.00**	72	198 (25%)	–	530	0.99–1.37
7	N	1.64**	114	13 (1%)	190 (15%)	570	1.33–2.43
8	N	1.26**	88	92 (15%)	–	560	1.62–1.82
9	A	2.50	91	15 (5%)	–	535	1.11–1.46
10	N	1.03**	64	14 (5%)	–	560	1.14–1.72
11	R	1.70	140	22 (6%)	125 (12%)	520	0.93–1.12
12	N	1.00	97	18 (8%)	–	510	0.79–0.82
13	R	2.10	209	25 (3%)	–	530	1.01–1.66
14	N	1.22	151	60 (3%)	–	530	1.60–1.92
15	N	1.30**	174	35 (9%)	–	520	0.67–0.91

Titanomagnetite oxidation state is determined by classifying at least 25 grains in each section. Grain sizes are the arithmetic means of all determinations and the volumes of sulphide and ilmenite are expressed as percentages of the total content of opaque minerals.

* N = normal; R = reversed; A = anomalous.

** Incipient development of maghemite in non-oxidised grains.

5. Discussion

Previous palaeomagnetic results from the coast-parallel dyke swarm have been reported by Ketelaar (1963) and Tarling (1966); these authors found reversed polarities while, in the regions studied here, the dykes are predominantly of normal polarity. Intrusion of the dyke swarm was spread over an interval of time spanning a polarity transition, and in the Tugtutôq region there is a systematic distribution of polarity (Fig. 1) from reversed at the seaward edge of the swarm (site *3*) through dykes intruded during a polarity transition (sites *4* and *5*) to normally magnetised dykes (remainder of the swarm in this region). This is concise evidence for a migration of the response to the regional stress field controlling intrusion of the dykes either seawards or landwards; migration of this effect took place sufficiently rapidly for dykes to be intruded within a discrete zone during a polarity transition (a few thousands of years).

Further west in the Arsuk region the picture is more complicated; the dykes are bunched together but both polarities and a transition are represented. This polarity transition may be the same as that represented further to the southeast (the dykes furthest inland are normal in both cases) but there is no positive evidence for lateral migration of the regional stress field in this area. Watt (1969) suggests that the S30°E and quartz-normative dykes in the west are slightly older than the east–west trending and olivine –nepheline-normative dykes in the south, and if this is the case the reversed dykes probably predate the normal dykes; migration of the tensile stress field controlling the dykes would then have been inland. McElhinny and Burek (1971) recognise three short reversal events in the Jurassic period with the single Mateke event (~168–170 m.y.) occurring in the Middle Jurassic (Bajocian Stage); this event probably correlates with the reversal identified here.

The palaeomagnetic pole position from the dykes at 190.9°E, 71.5°N is moved to 177.1°E, 77.4°N by closing the Davis Strait according to the 500-fathom continental reconstruction of Bullard et al. (1965) which rotates Greenland to Europe (22.0° anti-clockwise about an Euler pole at 73°N, 97.5°E, and Greenland–Europe to North America (38°, clockwise about on Euler pole at 88.5°N, 27.7°E). This is also equivalent to rotating Greenland clockwise by 18°, about an Euler pole at 70.5°N, 94.4°W.

Palaeomagnetic pole positions derived from Mesozoic rocks of Greenland include a pole from Upper Triassic sediments of eastern Greenland (Reeve et al., 1974) and two poles derived from small collections of Lower and Middle Triassic sediments of east Greenland (Athavale and Sharma, 1974). These pole positions are listed in Table III together with their positions after rotation to close the Davis Strait according the Bullard et al.'s (1965) fit, and rotated positions are plotted in Fig. 3 together with North American Triassic and Jurassic palaeomagnetic poles. The Triassic poles from Greenland all fail to agree with North American results of comparable age, although results (2) and (3) of Table III are closer to European data on the Bullard et al. (1965) reconstruction and palaeolatitudes of these latter two regions show general agreement (Athavale and Sharma, 1974; Reeve et al., 1974). These discrepancies have been explained in terms of large non-dipole components to the magnetic field (Briden et al., 1972), a 20° right lateral strike slip between

TABLE III

Palaeomagnetic poles from Mesozoic rocks of Greenland before and after correction for continental drift

Rock unit/age	Reference	Palaeomagnetic pole	
		before drift correction	after drift correction
(1) Pingodal Formation, L. Triassic	Athavale and Sharma (1974)	184°E, 34.5°N	166.9°E, 40.5°N
(2) Gipsdalen Formation, M. Triassic	Athavale and Sharma (1974)	158°E, 49°N	137.4°E, 54.2°N
(3) Fleming Fjord Formation, U. Triassic	Reeve et al. (1974)	103.2°E, 34°N	82°E, 34.7°N
(4) Coast-parallel dykes, M. Jurassic	this paper	190.9°E, 71.5°N	176.6°E, 77.5°N

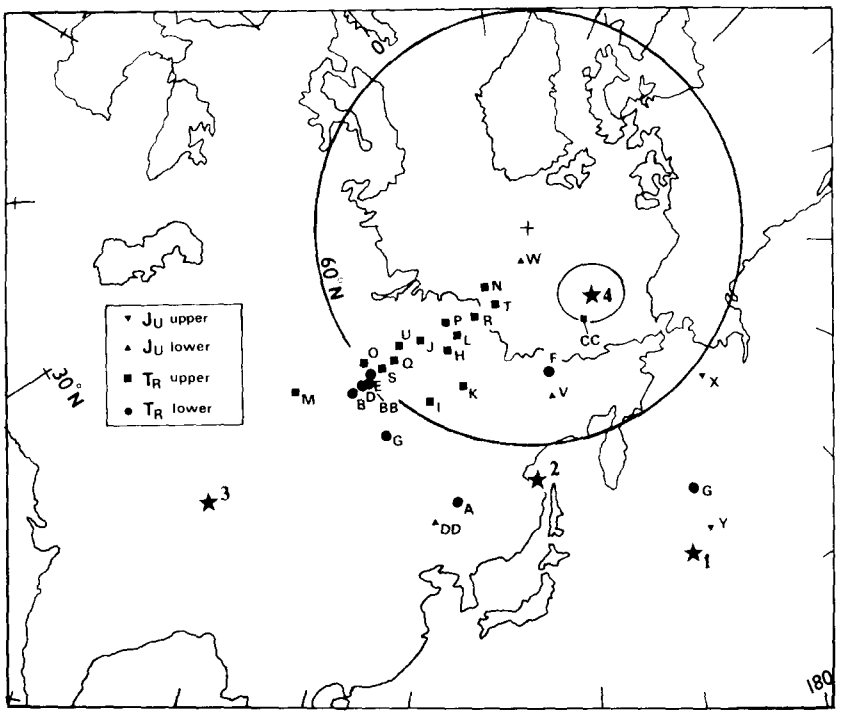


Fig. 3. Upper Triassic to Upper Jurassic poles from North America (small symbols) and Greenland (large stars, Table III) plotted on a Transverse Mercator projection centred on meridian 135°E. The poles from Greenland are plotted after closing the Davis Strait according to the reconstruction of Bullard et al. (1965). The North American poles are plotted from listings in the Geophysical Journal of the Royal Astronomical Society and given in the form list number/pole number are: *A* = 11/58; *B* = 11/59; *C* = 10/96; *D* = 9/51; *E* = 9/52; *F* = 9/53; *G* = 9/54; *H* = 8/68; *I* = 8/69; *J* = 9/50; *K* = 9/49; *L* = 10/89; *M* = 10/81; *N* = 10/90; *O* = 10/82; *P* = 10/83; *Q* = 10/84; *R* = 10/85; *S* = 10/86; *T* = 10/87; *U* = 10/88; *V* = 9/47; *W* = 8/61; *X* = 9/43; *Y* = 9/42. Additional poles are from Steiner and Helsley (1972, pole *BB*), Opdyke and Wensink (1966, sites *6–8*, pole *CC*) and Symons (1974, pole *DD*).

North America and Europe (Roy, 1972), and initial opening of the Atlantic during or prior to Triassic times (Athavale and Sharma, 1974). The scope of the latter two explanations is severely restricted by evidence for Middle Jurassic and later rifting between Greenland and Labrador (Clarke and Upton, 1971; Le Pichon et al., 1971), and by the correlation of Precambrian features across the Davis Strait (Bridgwater et al., 1973).

Jurassic palaeomagnetic poles from North America show much more scatter than Triassic results from the same region (Fig. 3) and are indicative of considerable apparent polar movement during this period. Because of this scatter the new Middle Jurassic pole from Greenland cannot be precisely compared with contemporary North American data, but it does lie within the scatter of North American Jurassic palaeomagnetic poles. A motion of Greenland, by 20° south as required by the second hypothesis outlined above would remove this pole from all but one of the North American Jurassic poles, and would appear to restrict such a motion to pre-Middle Jurassic time. Also, in view of the lack of consistency between Triassic palaeomagnetic results from Greenland, data for this period would appear to be in need of critical revaluation.

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